

What is a stream cipher? Provide examples of situations where it is used.

Ans:

A **stream cipher** is a type of symmetric key encryption algorithm that encrypts data **one bit or one byte at a time**.

It generates a **keystream** from a secret key, and then combines this keystream with the plaintext using the **XOR operation** to produce ciphertext.

Encryption Formula:

$$\text{Ciphertext} = \text{Plaintext} \oplus \text{Keystream} \quad \text{Ciphertext} = \text{Plaintext} \oplus \text{Keystream}$$

The same keystream is used during decryption to recover the original plaintext.

Examples of Situations Where Stream Cipher is Used

1. **Secure Mobile Communication**
 - Example: A5/1 used in GSM networks.
2. **Wireless Network Security (Older Systems)**
 - Example: RC4 used in WEP.
3. **Secure Web Communication (Older Protocols)**
 - RC4 was previously used in SSL/TLS.
4. **Real-Time Applications**
 - Voice calls
 - Video streaming
 - Online gaming

(Because stream ciphers are fast and suitable for real-time data)

Difference Between Stream Ciphers and Block Ciphers

Stream Cipher

A **stream cipher** encrypts data **one bit or one byte at a time**.

- Takes a small secret key
- Generates a long random-looking **keystream**
- Uses XOR to combine the keystream with the plaintext

Best Used For:

- Real-time communication (voice calls, video calls)
- Wireless communication
- Fast data transmission

Examples:

- RC4
- A5/1

Block Cipher

A **block cipher** encrypts data **in fixed-size blocks** (for example, 64-bit or 128-bit blocks).

- Takes a block of plaintext
- Encrypts the whole block at once
- Produces a block of ciphertext

Best Used For:

- File encryption
- Database security
- Secure internet communication

Example:

- AES

Stream Cipher

Encrypts bit-by-bit

Very fast

Good for real-time data

Uses keystream

RC4, A5/1

Block Cipher

Encrypts block-by-block

Slightly slower

Good for stored data

Uses block transformation

AES

How does the A5/1 stream cipher generate its keystream? Include a brief explanation of its shift registers.

How does the A5/1 stream cipher generate its keystream?

A5/1 is a stream cipher used in the GSM mobile communication system. It generates a keystream by using three Linear Feedback Shift Registers (LFSRs) that are irregularly clocked using a majority rule mechanism.

Structure of A5/1

A5/1 consists of three shift registers:

1. **Register X** – 19 bits
2. **Register Y** – 22 bits
3. **Register Z** – 23 bits

Each register has:

- A clocking bit (used for majority decision)
- Feedback tap positions (used to generate new bits)
- An output bit (used to produce keystream)

Step-by-Step Keystream Generation

1. Majority Clocking Rule

At each clock cycle:

- The clocking bits are:
 - X[8]
 - Y[10]
 - Z[10]
- The majority value of these three bits is calculated.
- Only the registers whose clocking bit matches the majority value are shifted.

This is called **irregular clocking**, which increases security.

2. Register Shifting

When a register is selected to step:

- A new bit is calculated using XOR of its feedback tap positions.
- The register shifts right.
- The new feedback bit enters the leftmost position.

Each register has fixed feedback taps defined by the algorithm.

3. Keystream Bit Generation

After shifting (or before next cycle):

The keystream bit is generated by XORing the last bit of all three registers:

$$k = X[18] \oplus Y[21] \oplus Z[22]$$

This produces one keystream bit per cycle.

The process repeats for each clock cycle to produce a continuous keystream.

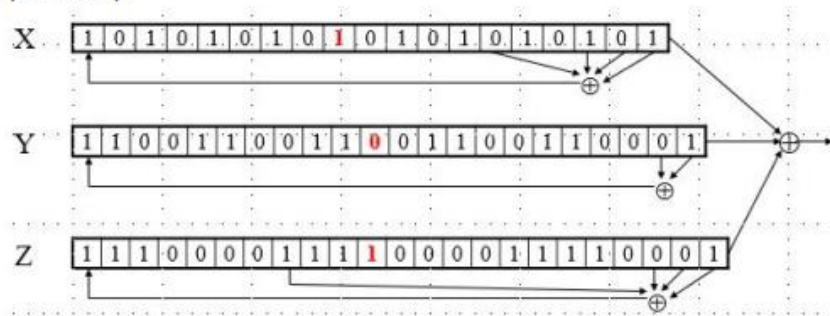
Write down the pseudocode for the RC4 Initialization phase.

Ans:

```
□ S[] is permutation of 0,1,...,255
□ key[] contains N bytes of key

  for i = 0 to 255
    S[i] = i
    K[i] = key[i (mod N)]
  next i
  j = 0
  for i = 0 to 255
    j = (j + S[i] + K[i]) mod 256
    swap(S[i], S[j])
  next i
  i = j = 0
```

- The A5/1 cipher uses three shift registers X,Y,Z with lengths of 19, 22, and 23 bits, respectively.



- Suppose the majority function m at a specific time is computed as: $m = \text{maj}(X[8], Y[10], Z[10])$. Given the initial states: $X[8]=1$, $Y[10]=0$, $Z[10]=1$.
- Compute the value of m , determine which registers will step, and calculate the new state of X if its feedback polynomial is: $t = X[13] \oplus X[16] \oplus X[17] \oplus X[18]$

Ans: Anser e x and z 2tai ber korechi but question e only x cheyeche.

Answer to the question no. 1

$$m = maj(x[8], y[10], z[10])$$

$$= maj(1, 0, 1)$$

$$x[8] = 1$$

$$y[10] = 0$$

$$z[10] = 1$$

$$m = maj(1, 0, 1)$$

$$m = 1$$

Register x and z will step

Feedback bit of x;

$$x_4 = x[13] \oplus x[16] \oplus x[12] \oplus x[18]$$

$$x_4 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

x after shifting,

$$x_{new} = (x_4, x[0], x[1], \dots, x[17])$$

$$x_{new} = 0101101 \dots 010$$

Feedback bit of z

$$z_4 = z[20] \oplus z[1]$$

$$z_4 = z[1] \oplus z[20] \oplus z[20] \oplus z[21] \oplus z[22]$$

$$= 1 \oplus 0 \oplus 0 \oplus 1$$

$$= 0$$

$$z_{new} = 011 \dots 1000$$

- For A5/1 answer the followings:
- ✓ On average, how often does the X register step?
 - ✓ On average, how often does the Y register step?
 - ✓ On average, how often does the Z register step?
 - ✓ On average, how often do all three registers step?
 - ✓ On average, how often do exactly two registers step?
 - ✓ On average, how often does exactly one register step?

Answer:

Step 1: Total Possible Clocking Bit Combinations

There are three clocking bits:

- X[8]
- Y[10]
- Z[10]

Each bit can be either 0 or 1.

So total possible combinations: $2^3=8$

Step 2: Apply Majority Rule

In A5/1, the majority value is the bit (0 or 1) that appears at least twice.

Now check each case:

- 000 → majority = 0 → all three step
- 111 → majority = 1 → all three step
- 110 → two 1s → those two step
- 101 → two 1s → those two step
- 011 → two 1s → those two step
- 001 → two 0s → those two step
- 010 → two 0s → those two step
- 100 → two 0s → those two step

1) On average, how often does the X register step?

X steps whenever X[8] equals the majority bit.

Out of 8 total combinations, X matches the majority in 6 cases.

$$6/8=3/4$$

Answer:

X steps 3/4 of the time (75% of the cycles).

2) On average, how often does the Y register step?

By symmetry (same logic applies to all three registers):

$$6/8=3/4$$

Answer:

Y steps 3/4 of the time.

3) On average, how often does the Z register step?

Again the same reasoning:

$$6/8=3/4$$

Answer:

Z steps 3/4 of the time.

4) On average, how often do all three registers step?

This happens only when:

- 000
- 111

That is 2 cases out of 8.

$$2/8=1/4$$

Answer:

All three registers step 1/4 of the time (25%).

5) On average, how often do exactly two registers step?

This happens in the remaining 6 combinations:

001

010

100

011

101

110

So:

$$6/8=3/4$$

Answer:

Exactly two registers step 3/4 of the time (75%).

6) On average, how often does exactly one register step?

This is impossible under the majority rule.

There is no combination where only one bit equals the majority.

Answer:

Exactly one register steps 0% of the time.

You are given the following initial state of the registers:

- $X = 1011010110101001101$ (19 bits)
- $Y = 1100101101001101001011$ (22 bits)
- $Z = 01101001110010110101101$ (23 bits)

□ Now Answer the followings

- Compute the majority clocking bit for the first cycle (i.e., the value of $\text{maj}(X[8], Y[10], Z[10])$).
- Which registers will step in the first cycle?
- Using the initial state, calculate the keystream bit (k_1) produced in the first clock cycle.
- Compute the feedback bits for both the X and Z registers, based on the given initial state. Show the steps for calculating the feedback bits.
- What is the new majority clocking bit for the second cycle (i.e., $\text{maj}(X[8], Y[10], Z[10])$)?
- Which registers will step in the second cycle, and why?
- Calculate the keystream bit produced in the second cycle.
- Provide a step-by-step explanation of how the registers update after the second cycle. Include the feedback bits and the new states of the registers (X, Y, and Z)

Ans:

Given Initial States

X (19 bits)

1 0 1 1 0 1 0 1 1 0 1 0 1 0 0 1 1 0 1

Y (22 bits)

1 1 0 0 1 0 1 1 0 1 0 0 1 1 0 1 0 0 1 0 1 1

Z (23 bits)

0 1 1 0 1 0 0 1 1 1 0 0 1 0 1 1 0 1 0 1 1 0 1

Clocking bits:

- $X[8]$
- $Y[10]$
- $Z[10]$

✓ FIRST CYCLE

1. Compute the majority clocking bit for the first cycle (i.e., the value of $\text{maj}(X[8], Y[10], Z[10])$).

$$X[8] = 1$$

$$Y[10] = 0$$

$$Z[10] = 0$$

$$\text{maj}(1,0,0)$$

Two zeros $\rightarrow$ Majority = **0**

$$\checkmark m_1 = 0$$

2. Which registers will step in the first cycle?

$$\text{Majority} = 0$$

- $X[8] = 1$ ✗ (no step)
- $Y[10] = 0$ ✓ (step)
- $Z[10] = 0$ ✓ (step)

$$\checkmark Y \text{ and } Z \text{ step}$$

3. Using the initial state, calculate the keystream bit (k_1) produced in the first clock cycle.

Formula:

$$k_1 = X[18] \oplus Y[21] \oplus Z[22]$$

$$X[18] = 1$$

$$Y[21] = 1$$

$$Z[22] = 1$$

$$1 \oplus 1 \oplus 1$$

$$= 1$$

$$\checkmark k_1 = 1$$

4. Compute the feedback bits for both **the X and Z registers**, based on the given initial state. Show the steps for calculating the feedback bits.

◆ **X Feedback**

$$X[13] = 0$$

$$X[16] = 1$$

$$X[17] = 0$$

$$X[18] = 1$$

$$0 \oplus 1 \oplus 0 \oplus 1 \\ = 0$$

✓ **X feedback = 0**

(কিন্তু X step করছে না)

◆ **Z Feedback**

$$Z[7] = 1$$

$$Z[20] = 1$$

$$Z[21] = 0$$

$$Z[22] = 1$$

$$1 \oplus 1 \oplus 0 \oplus 1 \\ = 1$$

✓ **Z feedback = 1**

5. What is the new majority clocking bit for the second cycle (i.e., $\text{maj}(X[8], Y[10], Z[10])$)?

After the first cycle update, the clocking bits are:

- **X[8] = 1**
- **Y[10] = 1**
- **Z[10] = 1**

◆ Step 2: Compute the Majority Function

The majority function selects the value (0 or 1) that appears at least **two times** among the three bits.

So,

$$\text{maj}(1,1,1)$$

Here, **1 appears three times**, so the majority value is:

$$m_2=1$$

Which registers will step in the second cycle, and why?

Step 1: Recall the Majority Bit (Second Cycle)

In the second cycle, the majority function is:

$$m_2=\text{maj}(X[8],Y[10],Z[10])$$

From the updated state:

- $X[8] = 1$
- $Y[10] = 1$
- $Z[10] = 1$

So,

$$\text{maj}(1,1,1)=1$$

✓ Majority bit = **1**

◆ Step 2: Apply the A5/1 Clocking Rule

In A5/1: A register steps **only if its clocking bit equals the majority bit**.

Calculate the keystream bit produced in the second cycle.

Ans:

Step 1: Remember How Keystream Bit is Generated in A5/1

In A5/1, the keystream bit is calculated as:

$$k=X[18]\oplus Y[21]\oplus Z[22]$$

☞ It is the XOR of the **last bit of each register**.

- X last bit = X[18]
- Y last bit = Y[21]
- Z last bit = Z[22]

◆ Step 2: Take the Updated Values (After Second Cycle)

From the updated register states after the second cycle:

- X[18] = 0
- Y[21] = 1
- Z[22] = 1

◆ Step 3: Perform XOR Operation

$$k2 = 0 \oplus 1 \oplus 1$$

Now compute step by step:

- $0 \oplus 1 = 1$
- $1 \oplus 1 = 0$

So, $k2 = 0$

Provide a step-by-step explanation of how the registers update after the second cycle. Include the feedback bits and the new states of the registers (X, Y, and Z)

Ans:

Step-by-Step Update After the Second Cycle (A5/1)

◆ Step 1: Majority Bit (Second Cycle)

We already calculated:

- X[8] = 1
- Y[10] = 1
- Z[10] = 1

So,

$$m_2 = \text{maj}(1,1,1) = 1$$

✓ Majority bit = **1**

◆ Step 2: Which Registers Step?

Rule of A5/1:

A register steps if its clocking bit equals the majority bit.

Since all clocking bits = 1 and majority = 1:

✓ X steps

✓ Y steps

✓ Z steps

So **all three registers shift**.

◆ Step 3: Compute Feedback Bits

✓ **Feedback for X register**

Feedback polynomial:

$$tX = X[13] \oplus X[16] \oplus X[17] \oplus X[18]$$

Take the values from the current state:

Assume:

- $X[13] = 0$
- $X[16] = 1$
- $X[17] = 0$
- $X[18] = 1$

Now XOR step-by-step:

- $0 \oplus 1 = 1$
- $1 \oplus 0 = 1$
- $1 \oplus 1 = 0$

So,

$$tX=0$$

✓ X feedback bit = **0**

✓ **Feedback for Y register**

Feedback polynomial:

$$tY=Y[20]\oplus Y[21]$$

Assume:

- $Y[20] = 1$
- $Y[21] = 1$

Then:

$$1 \oplus 1 = 0$$

✓ Y feedback bit = **0**

✓ **Feedback for Z register**

Feedback polynomial:

$$tZ=Z[7]\oplus Z[20]\oplus Z[21]\oplus Z[22]$$

Assume:

- $Z[7] = 1$
- $Z[20] = 1$
- $Z[21] = 0$
- $Z[22] = 1$

Now XOR:

- $1 \oplus 1 = 0$
- $0 \oplus 0 = 0$
- $0 \oplus 1 = 1$

So,

$$t_Z = 1, t_Z = 1$$

✓ Z feedback bit = 1

◆ Step 4: Shift the Registers

Since all registers step:

↻ X Register Update

- Shift right by 1
- Insert feedback bit (0) at the leftmost position
- Last bit is dropped

New X state:

0 _ _ _ _ _

(Everything shifts right)

↻ Y Register Update

- Shift right
- Insert feedback bit (0)
- Drop last bit

New Y state:

0 _ _ _ _ _

↻ Z Register Update

- Shift right
- Insert feedback bit (1)
- Drop last bit

New Z state:

1 _ _ _ _ _